OXIDATION NUMBER EXCEPTIONS – COMPLETE CHEATSHEET (NEET | JEE | BOARDS)

1. Oxygen Exceptions

Normal: O = –2

Peroxides (H■O■, Na■O■, BaO■) → O = –1

Superoxides (KO■, RbO■) → O = –½

Ozonides (KO■) → O = –■

OF■ → O = +2

O■F■ → O = +1

Elemental O■, O■ → O = 0

2. Hydrogen Exceptions

Normal: H = +1

Metal Hydrides (NaH, CaH■, NaBH■, LiAlH■) → H = –1

3. Halogen Exceptions

Normal: Halogens = –1

In Oxoacids (HClO, HClO■, HClO■, HClO■) → Cl = +1, +3, +5, +7

In Interhalogens (ICl, IF■, BrF■) → Larger halogen is positive

Fluorine is always –1

4. Most Important Exam Compounds

KMnO■ → Mn = +7

K■Cr■O■ → Cr = +6

H■O■ → O = –1

CrO■ → Cr = +6 (contains peroxo O)

H■SO■ → S = +6 (peroxo acid)

H■S■O■ → S = +6

Na■S■O■ → S = –2 and +6

NH■NO■ → N = –3 and +5

5. Mixed Oxidation State Compounds

Fe■O■ → Fe = +2 and +3

Mn■O■ → Mn = +2 and +3

6. Nitrogen Special Cases

N■O → N = 0 and +2 (average +1)

NO■ → N = +4 (resonance mixture of +3, +5)

7. Sulphur Special Cases

SO■ → S = +4

SO■ → S = +6

Na■S■O■ → S = –2 and +6

8. Noble Gas & Special JEE Compounds

XeO■ → Xe = +6

$XeO_4 \rightarrow Xe = +8$

$XeF_4 \rightarrow Xe = +4$

$XeF_6 \rightarrow Xe = +6$

$Fe(CO)_5 \rightarrow Fe = 0$

$CH_3MgBr \rightarrow C = -3$

Quick Revision Table (Most Repeated NEET/JEE Exceptions)

Compound	Key Exception	Oxidation Number
H_2O_2	Peroxide	O = –1
KO_2	Superoxide	O = –½
OF_2	Fluorine effect	O = +2
CrO_5	Peroxo oxygen	Cr = +6
H_2SO_5	Peroxo acid	S = +6
$Na_2S_2O_3$	Two S atoms	–2 and +6
NH_4NO_3	Two N atoms	–3 and +5
Fe_3O_4	Mixed oxide	+2 and +3